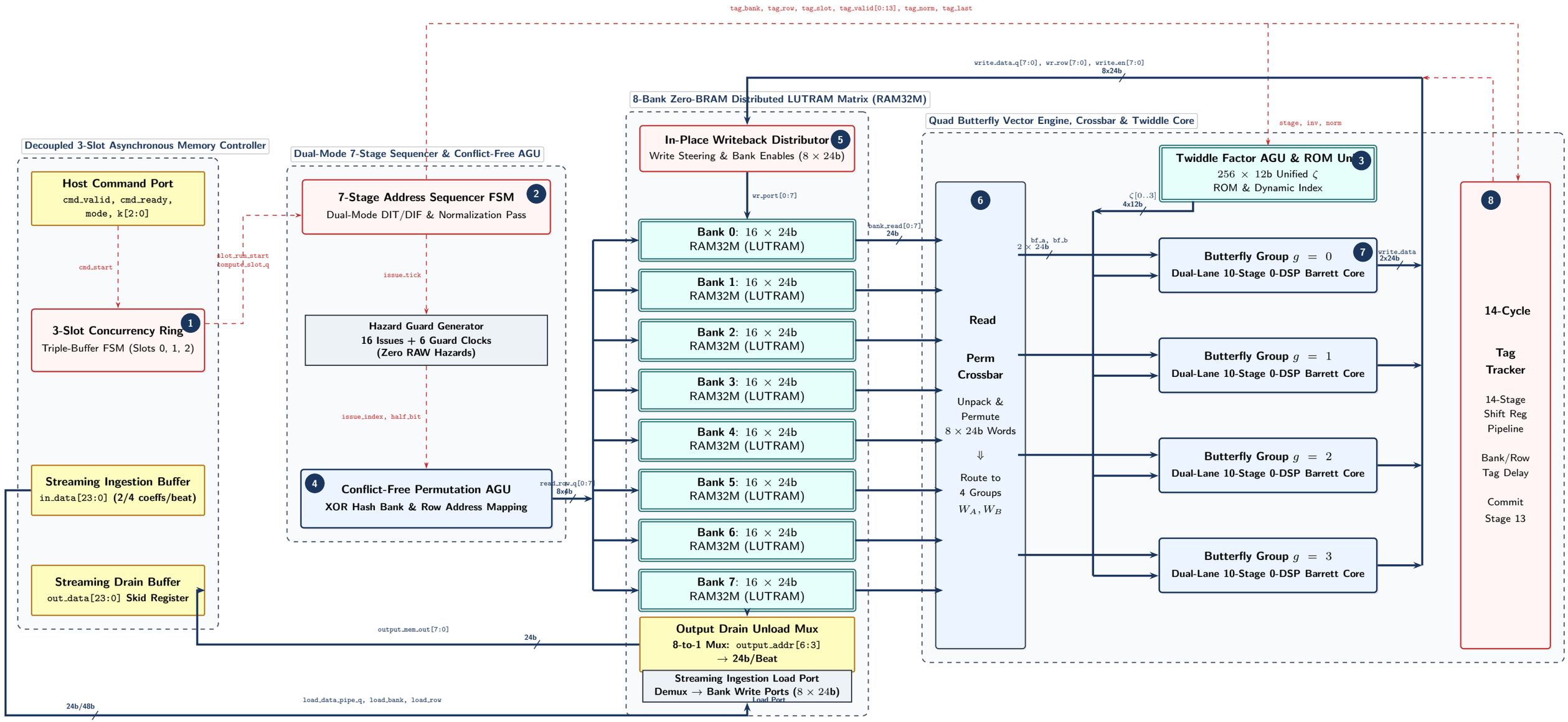
ML-KEM HPKA: Subsystem 04 — NTT/INTT 8-Bank Memory Architecture & Address Sequencer

Conflict-Free Distributed LUTRAM Matrix, Dual-Mode 7-Stage Sequencer, & Pipelined Butterfly Interconnect — Module: `mlkem_ntt_intt.sv`



ID	Component / Signal	Width / Format	Description
1	3-Slot Concurrency Ring	Control / 3-State FSMs	Manages concurrent loading, compute, and output draining across 3 polynomial slots without bubbles.
2	7-Stage Address Sequencer	9-bit Counter / Control	Schedules forward DIT ($h = 6 - s$), inverse DIF ($h = s$), and normalization, inserting 6 guard clocks per stage.
3	Twiddle AGU & ROM Unit	7-bit Addr / 4 × 12b Data	Generates bit-reversed twiddle addresses for the 256-entry ζ ROM and forces $\zeta = 3303$ during normalization.
4	Conflict-Free Permutation AGU	Combinational / 3-bit Bank	Maps addresses via XOR-folded hashing $\{(\bigoplus_{k=2}^{6} \text{addr}[k]), \text{addr}[1:0]\}$ to guarantee 8 contention-free reads per clock.
5	In-Place Writeback Distributor	8-bit Enables / 24-bit Words	Steers 8-word butterfly results and commit addresses back to destination banks without memory hazards.
6	Read Permutation Crossbar	8 × 24b → 8 × 12b Operands	Permutes bank outputs and unpacks 24-bit words into 16 coefficients bundled with twiddles for the 4 butterfly groups.
7	Quad Butterfly Processing Array	8 Scalar ALUs (8 × 12b)	Four dual-lane vector groups executing 10-stage zero-DSP Cooley-Tukey/Gentleman-Sande transforms with Barrett mod 3329.
8	14-Cycle Tag Pipeline Tracker	14-Stage Shift Register	Tracks bank/row tags in lockstep with the 14-cycle datapath to steer in-place writebacks without memory hazards.